

Setting up environment

Setting up a virtual security lab is a rite of passage. You're building a controlled "sandbox" where you can monitor traffic without risking your actual home network.

Since you have a mix of a pre-built VM (Kali) and raw ISOs, the process varies slightly for each. Here is your step-by-step guide to building this environment.

1. Global VirtualBox Networking Setup

Before touching the VMs, we need to ensure the **Host-Only** network exists.

1. Open VirtualBox.
 2. Click **File > Tools > Network Manager**.
 3. Under the **Host-only Networks** tab, click **Create**.
 4. Ensure a new adapter (usually vboxnet0 or VirtualBox Host-Only Ethernet Adapter) appears.
 5. **Uncheck** the "DHCP Server" box. Since you are using static IPs (10.0.0.x), we don't want VirtualBox assigning random IPs.
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2. Importing the Kali Linux VM

Since your Kali is a downloaded VM (likely an .ova or .ovf file):

1. Click **File > Import Appliance**.
 2. Select your Kali file and click **Next**.
 3. In the settings list, look for **MAC Address Policy** and select "Generate new MAC addresses for all network adapters" (this prevents conflicts).
 4. Click **Import**.
 5. **Configure Networking:**
 - Right-click the Kali VM > **Settings > Network**.
 - **Adapter 1:** Attached to: **NAT**.
 - **Adapter 2:** Check "Enable Network Adapter". Attached to: **Host-only Adapter**. Select the name created in Step 1.
 - Click **Advanced:** Set **Promiscuous Mode** to **Allow All**.
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3. Creating the ISO-Based VMs (Windows 7 & Ubuntu)

Follow these steps for the Windows 7 VM and both Ubuntu VMs (Snort & Web Server).

1. Click **New**.
 2. **Name & OS:** * *Win7*: Type "Windows 7", Version "Windows 7 (64-bit)".
 - *Ubuntu*: Type "Ubuntu Snort" or "Ubuntu Web", Version "Ubuntu (64-bit)".
 3. **ISO Image:** Click the dropdown and select your respective ISO file.
 4. **Hardware:** Allocate at least 2GB RAM for Ubuntu and 4GB for Windows 7.
 5. **Network (Crucial):**
 - Go to the **Network** tab in the creation wizard (or Settings after creation).
 - **Adapter 1:** Attached to **NAT**.
 - **Adapter 2:** Enable and attach to **Host-only Adapter**.
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4. Guest OS Configuration (Static IP Assignment)

Once the OSs are installed, you must manually set the internal IPs on the **second** interface.

Windows 7 (10.0.0.10)

1. Go to **Control Panel > Network and Sharing Center > Change adapter settings**.
2. Find the adapter labeled "Unidentified Network" (the second one).
3. Right-click > **Properties > IPv4 > Properties**.
4. Select **Use the following IP address**:
 - **IP:** 10.0.0.10
 - **Subnet Mask:** 255.255.255.0
 - Leave Gateway/DNS blank (the NAT adapter handles internet).

Ubuntu & Kali (10.0.0.11, .12, .13)

Open a terminal and edit the Netplan config (Ubuntu) or use Network Manager (Kali/Ubuntu Desktop). For terminal-based Ubuntu:

1. Run `sudo nano /etc/netplan/01-netcfg.yaml`.
 2. Configure the second interface (e.g., `enp0s8`):
YAML

```
network:
  version: 2
  ethernets:
    enp0s8:
      addresses: [10.0.0.x/24] # Replace x with .11, .12, or .13
```
 3. Run `sudo netplan apply`.
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5. Configuring MSF

Once the Kali is installed, you must manually create the MSF Database.

1. Login with **Username** *kali* and **Password** *kali*
2. Open a new **Terminal**
3. Create the MSF Database
`sudo msfdb init`

6. Summary Table of Lab Topology

VM Name	Role	Adapter 1 (Internet)	Adapter 2 (Internal)	Promiscuous
Windows 7	Target/Client	NAT	10.0.0.10	Deny
Ubuntu Web	Target/Server	NAT	10.0.0.11	Deny
Kali Linux	Attacker	NAT	10.0.0.12	Allow All
Ubuntu Snort	IDS/Sensor	NAT	10.0.0.13	Allow All

6. Verifying the Setup

To ensure your "Security Lab" is actually talking internally:

1. From Kali, try to ping the Web Server: ping 10.0.0.11.
2. If it fails, check if the Windows/Ubuntu firewalls are blocking ICMP (Ping).
3. **Note on Snort:** Since Snort is on a Host-Only network, it will only "see" traffic directed to it unless you use a tool like arpspoof from Kali to force traffic through the Snort VM, or configure a bridge.

Would you like me to provide the specific Snort commands to start sniffing traffic on that 10.0.0.x interface?